

Binary Payment Schemes: Moral Hazard and Loss Aversion

Herweg, Müller, and Weinschenk 2010
Experimental Design

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Outline

- 1 Motivation
- 2 Related Literature
- 3 Experiment Design
- 4 Analyses

Summary: Why Binary Schemes?

- **Classical Contract Theory** (Holmström 1979): Optimal contracts should be **complex**, exploiting all available information.
- **Observation:** Real-world contracts are often remarkably **simple** (e.g., binary bonus schemes, flat wages).

Herweg, Müller, and Weinschenk (2010) address this puzzle by introducing **expectation-based loss aversion** into the standard principal-agent model with moral hazard, building on Kőszegi and Rabin (2007). They identify a trade-off of two opposing effects of adding wage levels:

- 1 **Incentive Gain:** More wage levels (complexity) provide better incentives.
- 2 **Loss Aversion Cost:** Every additional wage level creates more possible unfavourable comparisons, raising expected psychological costs.

From Theory to Experiment: Research Question

Motivation: Why Binary Schemes?

The central theoretical prediction by Herweg, Müller, and Weinschenk (2010) that **binary payment schemes are optimal** has not yet been directly tested empirically.

Do binary bonus contracts increase effort beyond multi-level contracts, and is this effect driven by agents' expectation-based loss aversion?

Related Empirical Literature

- **Abeler et al. (2011)**: Conducted a laboratory experiment showing that expectation-based reference points directly drive workers' effort provision.
- Abeler et al. (2011) prove the existence of expectation-based reference points in effort provision; my experiment **explicitly tests the contractual solution (Binary vs. Multi-level)** proposed by Herweg, Müller, and Weinschenk (2010).
- Labour Supply (Cab Drivers): **Crawford and Meng (2011)** shows that a model with expectation-based targets significantly outperforms the neoclassical benchmark in predicting cab drivers' effort.

Hypotheses

The hypotheses are derived directly from the theoretical predictions in Herweg, Müller, and Weinschenk (2010) and the expectation-based reference-point model of Köszegi and Rabin (2007).

- ➊ **H1 (Main Effect Contract):** Effort under the binary contract is greater than or equal to effort under the multi-level contract.
→ *Mechanism: Binary contracts minimize the expected sensation of loss.*
- ➋ **H2 (Main Effect Reference Point):** A higher reference point leads to higher effort within a given contract type.
→ *Mechanism: Agents adjust their effort to avoid falling short of their expectation-based reference point.*
- ➌ **H3 (Interaction Effect):** The effort difference between reference-point conditions is *larger* under the multi-level contract than under the binary contract.
→ *This is the central finding of our design. The interaction term ($\beta_3 < 0$).*

Sampling Methodology & Recruitment

- **Target Population & Subject Pool:**

- ▶ Undergraduate and graduate students across various disciplines.
- ▶ Recruitment via the university's standard subject pool system (e.g., ORSEE at the Cologne Laboratory for Economic Research).

- **Sample Size (N):**

- ▶ Total $N = 252$ participants (63 per treatment group).
- ▶ Pre-determined via a priori power analysis ($\alpha = 0.05$, $1 - \kappa = 0.80$) to detect a medium effect size **for pairwise cell comparisons**.

- **Randomisation & Assignment:**

- ▶ Pure between-subject design.
- ▶ Participants are randomly assigned to one of the 4 experimental conditions.

- **Incentivise & Execution:**

- ▶ *Performance-based pay* to ensure salient economic incentives.
- ▶ Base show-up fee (e.g., 2.5€) to minimise selection bias and attrition.

Structure of Experiment

Experimental Flow & 2×2 Between-Subject Design

Instructions & Control Questions (~ 8 min)

General rules, slider task tutorial & comprehension test



Round 0: Baseline Task (~ 3 min)

Unrewarded measurement of individual ability a_i



Round 1: Main Task (Randomized to 1 of 4 Groups)

	Low Reference Point	High Reference Point
Binary Contract	G1 (Binary + Low)	G2 (Binary + High)
Multi-level Contract	G3 (Multi-level + Low)	G4 (Multi-level + High)



Post-Experimental Questionnaire (~ 10 min)

Loss Aversion (λ_i), Risk Preference (ρ_i), Checks & Demographics



Payout / Incentive Realization (~ 5 min)

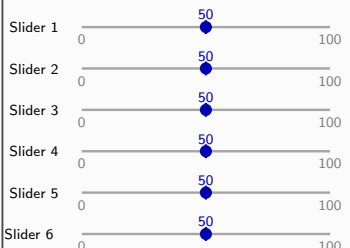
Show-up fee + Earnings from Round 1

The Real Effort Task: Slider Task

Experiment Design, by Gill and Prowse (2019)

What participants do

Move each slider to position 50
48 sliders in total



Slider 1 0 50 100

Slider 2 0 50 100

Slider 3 0 50 100

Slider 4 0 50 100

Slider 5 0 50 100

Slider 6 0 50 100

Time limit: 120 seconds per round

Real-time feedback

✓ Slider 3 is correct

✓ Slider 7 is correct

! Slider 9 is incorrect



Participants can correct mistakes within the 120-second window

Outcome:
Number of sliders correct

Round 0: Baseline Task

Experiment Design

Problem: We need to find a way to introduce a payment structure which is based on individual performance.

Solution: Unpaid slider task to determine individual ability a_i .

- We can ensure all participants face the same 80% success probability.

$$\hat{e}_i = 0.8 \times a_i$$

- Loss aversion *uniformly* across all groups.
- Fair and realistic comparison: Effort differences reflect behavioural response, not ability differences.

Round 1: 2×2 Between-Subject Design

Experiment Design

	Low RP	High RP
Binary Contract	Group 1	Group 2
Multi-level Contract	Group 3	Group 4

Contract Structures:

- Binary: $w \in \{w_l, w_h\} = \{2\text{€}, 10\text{€}\}$
- Multi-level: $w \in \{w_l, w_m, w_h\} = \{2\text{€}, 6\text{€}, 10\text{€}\}$

$N = 252$ (63 per cell), pure between-subject.

Reference Point Manipulation (Abeler et al. 2011)

Experiment Design

Payout mechanism: After the slider task, a **50/50 coin toss** determines which payment counts:

- **Heads:** Performance payment (Binary or Multi-level contract)
- **Tails:** Fixed payment F

Low RP: $F_{Low} = 4\text{€}$

High RP: $F_{High} = 8\text{€}$

Why this isolates the reference point

The marginal incentive to exert effort (conditional on the performance branch) is **identical** across conditions. Only the reference point shifts, via F , **not** via the probability.

Payment Structure

Experiment Design

Groups 1 & 2: Binary Scheme

Sliders correct	Payment
$e_i < 0.8 a_i$	$w_l = 2$
$e_i \geq 0.8 a_i$	$w_h = 10$

Groups 3 & 4: Multi-Level Scheme

Sliders correct	Payment
$e_i < 0.5 a_i$	$w_l = 2$
$0.5 a_i \leq e_i < 0.8 a_i$	$w_m = 6$
$0.8 a_i \leq e_i$	$w_h = 10$

Realisation (identical for all groups): a fair coin flip decides the payout.

Heads ($p = 0.5$) Payout = performance pay $w \in \{w_l, w_m, w_h\}$

Tails ($p = 0.5$) Payout = fixed amount F

Note: By construction, the multi-level scheme has a (weakly) higher expected payoff at benchmark effort; potential confound between contract type and expected pay

Post-Experimental Survey & Measurement of Key Variables

Experiment Design

- **Loss Aversion (λ_i) [Primary Behavioral Variable]:**
 - ▶ Incentive-compatible Lottery Choice Task (Gächter et al., 2007)
 - ▶ Elicits individual loss aversion parameter λ_i via 6 coin-toss decisions
- **Risk Preference (ρ_i) [Control Variable]:**
 - ▶ General Risk Attitude (SOEP 0–10 Scale) & Financial Risk Domain
- **Demographics & Controls (X_i):**
 - ▶ Age, Gender, Field of Study, Math / Cognitive Ability Self-Assessment

Data Analysis Plan: Econometric Models

Testing Hypotheses & Research Questions

Model 1: 2x2 Factorial Design (Testing H1, H2, H3)

$$e_i = \beta_0 + \beta_1 \text{Binary}_i + \beta_2 \text{HighRP}_i + \beta_3 (\text{Binary}_i \times \text{HighRP}_i) + \gamma a_i + \delta X_i + \varepsilon_i$$

- ① **H1:** $\beta_1 \geq 0$ (*Binary contract yields equal or greater effort than multi-level*)
- ② **H2:** $\beta_2 > 0$ (*High reference point strictly increases effort*)
- ③ **H3:** $\beta_3 < 0$ (*Interaction: The High RP effect is smaller under the Binary contract*)
- (RCT) no need of the individual personality traits of the participants ($i; p_i$)

Data Analysis Plan: Econometric Models

Testing Research Questions

RQ Do binary bonus contracts increase effort beyond multi-level contracts, and is this effect driven by agents' expectation-based loss aversion?

Model 2: The Driver of the Effect (Addressing the RQ)

$$e_i = \alpha_0 + \alpha_1 \text{Binary}_i + \alpha_2 \lambda_i + \alpha_3 (\text{Binary}_i \times \lambda_i) + \delta X_i + \varepsilon_i$$

- λ_i : Individual degree of loss aversion (measured separately).
- $\alpha_3 > 0$ verifies the RQ: The effort-increasing effect of binary contracts is driven by agents with stronger loss aversion. (Tests whether the contract

Discussion: Design Choice: Real-Effort Slider Task

No experiment is perfect

Brüggen and Strobel (2007): "Real Effort" and "Chosen Effort" produce different effect sizes.

Our Consistency: We focus on the **Slider-Task** by Gill and Prowse (2019), because,

- ➊ **External Validity:** Realistic work behavior (relevant for practice)
- ➋ Herweg, Müller, and Weinschenk (2010) is interested in *real* Decisions based on effort, not stylized choices
- ➌ **Mechanism Clarity:** Loss aversion affects actual productivity, not just the psychology of choice.

Limitations:

- Ability-Confounder

Critical Discussion: Limitations & Trade-offs

No experiment is perfect

- Baseline Task Round 0:
 - ▶ Economic incentive to exert high effort?
 - ▶ Participants might intentionally slack off in Round 0 to secure an artificially low
- Deception and Rational Expectations:
 - ▶ The lottery avoids deception but may feel artificial compared to real-world peer comparisons.
 - ▶ **Precedent:** Abeler et al. (2011) use identical logic.
- Corner Solutions in Effort Provision:
 - ▶ Marginal cost of physical effort and fatigue are practically negligible
 - ▶ Is the Bonus of up to 8EUR even an incentive to perform?
 - ▶ Is there a use case of a control group (with no payment)?
- Disentangling Risk Aversion from Loss Aversion:
 - ▶ Intermediate wage (w_m) provides partial insurance and reduces income variance
 - ▶ Challenge to conclusively prove that the observed divergence in effort is strictly driven by expectation-based loss aversion rather than standard risk aversion

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Appendix: Power Calculations (2x2 Design)

Objective: Calculate required sample size using the course formula, accounting for 4 experimental groups.

Formula used:

$$N_{comp} = (t_{1-\kappa} + t_{\frac{\alpha}{2}})^2 \cdot \frac{\sigma^2}{P(1-P)\text{MDE}^2}$$

Parameters for a medium effect:

- $\alpha = 0.05 \implies t_{\alpha/2} \approx 1.96$
- $\kappa = 0.20$ (80% Power) $\implies t_{1-\kappa} \approx 0.84$
- $P = 0.5 \implies P(1-P) = 0.25$ (Maximizes power to minimize required N)
- $\text{MDE} = 0.5\sigma$ (Medium effect size)

Results & 4-Group Application:

- **Required N per pairwise comparison:** ≈ 126 (63 per group)
- **Planned Total N :** 252 (4×63 participants)
- **Justification:** By targeting $N = 252$ (63 per cell), the experiment is fully powered (80%) not only for main effects, but also for the most demanding **pairwise cell-to-cell comparisons** to detect an effect size of $\text{MDE} = 0.5\sigma$.

Appendix: Variable and Parameter Definitions

Symbol	Name	Definition / Type
Dependent Variables		
e_i	Effort	Measured effort of participant i (e.g., number of solved tasks)
BonusCosts $_i$	Incentive Costs	Total bonus payout made to participant i (in Experimental Currency Units)
Independent Variables (Main Treatments & Traits)		
Binary $_i$	Contract Treatment	Dummy: 1 if Binary Contract, 0 if Multi-level Contract
HighRP $_i$	Reference Point Treatment	Dummy: 1 if High Reference Point group, 0 if Low Reference Point group
λ_i	Loss Aversion	Continuous measure of individual expectation-based loss aversion (elicited separately)
Coefficients & Controls		
$\beta_k, \alpha_k, \theta_k$	Regression Coefficients	Estimated effects of the respective variables on the dependent variable
X_i	Control Variables	Vector of individual characteristics (e.g., age, gender, general risk aversion)
ε_i	Error Term	Unobserved, robust standard errors clustered at the session level

Appendix: between-subject problems

Problems with potential solutions

- Low Statistical Power: Use multiple Rounds per Person and Covariates
- No intra-individual comparison: (RCT) Randomisation ensures causality

Why the reference point would be contaminated in the “within” case.

- The reference point is not the status quo, but rather the rational expectation regarding one's own wage distribution. So if they did both